

11-07-2026

Agenda: (ML-17)

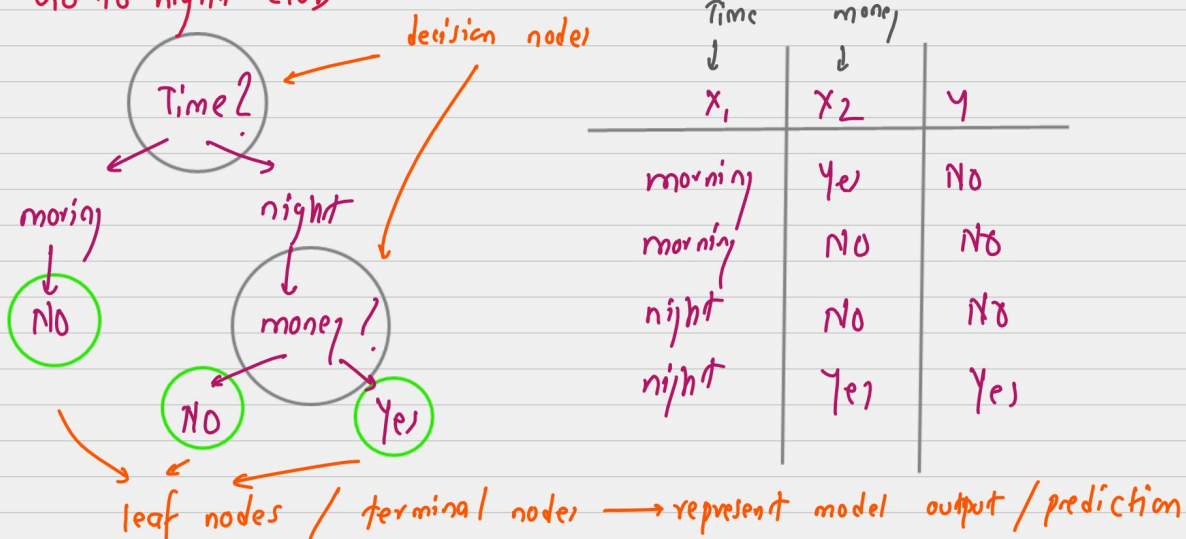
- Decision Tree (classification & Regression) (concepts)
- kmeans clustering (practical (wcss))

Decision Tree: —

DT is a supervised ML algorithm that breaks down a dataset into smaller subsets by asking a series of targeted, conditional "if-then-else" questions.

- supervised ML algo
- ease of interpretation
- both regression & classification

Task: Go to night club



Decisions navigated via conditional control statement

if-else statement → why it is a ML algo?

$x_1 \ x_2 \ x_3 \ x_4 \dots x_{100} \ y$ → root node?
 → next decision node?
 → too complex
 → ... other
 million of rows
 regression

→ DT
 → Random forest
 → Gradient Boosting (GB)
 → XGBoost
 → majority → regression
 → classification

Decision Tree classifier

Dataset:

weather	Temperature	play-Tennis
Sunny	Hot	NO
Sunny	Mild	NO
cloudy	Hot	Yes
Rain	Mild	Yes
Rain	Cool	Yes
cloudy	Cool	Yes
Rain	Hot	NO
Sunny	Cool	Yes

ML → linear Regression
 → gradient Descent
 → loss function / cost fn
 → regularization
 → w & b
 → $1/r$

ML - Decision Tree (classification)

→ purity → Impurity
 ↳ Entropy
 ↳ Information gain
 $IG(\text{parent}) - IG(\text{child})$

↳ Gini impurity
 ↳ Gini gain
 $G(\text{parent}) - G(\text{child})$

Impurity

$x_1 \ y$
 1 0.2 Yes
 2 0.3 Yes → if-else
 3 0.01 Yes
 4 0.8 Yes
 5 0.6 NO
 6 0.7 NO
 7 0.9 NO
 8 0.10 NO

if $x < 0.6$
Yes

else

NO

Decision node
 $x < 0.6$

Yes

NO

0.2 Yes
 0.3 Yes
 0.01 Yes
 → pure node

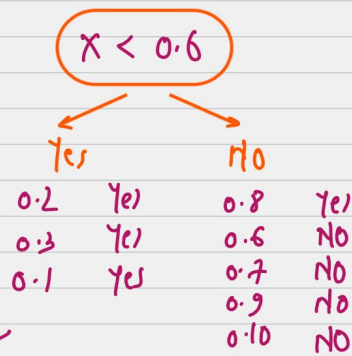
0.8 Yes
 0.6 NO
 0.7 NO
 0.9 NO
 0.10 NO
 ← impure node

pure node → all samples
 belong to same class

an impure node → mixed class

Entropy $\rightarrow$ measurement of uncertainty
 $-\sum p_i \log_2(p_i)$

Gini impurity $\rightarrow$ impurity
 $1 - \sum p_i^2$



Entropy

(Q1)	count	proportion
Yes	3	1

$$= -\sum p_i \log_2(p_i)$$

$$= -(1 \log_2(1))$$

$$\text{impurity} = 0$$

0-1

(Q1)	count	proportion
Yes	1	0.2
No	4	0.8

$$= -(0.2 \log_2(0.2) + 0.8 \log_2(0.8))$$

$$= 0.722$$

worst condition

(Q1)	count	proportion
Yes	4	0.5
No	4	0.5

$$= -(0.5 \log_2(0.5) + 0.5 \log_2(0.5))$$

$$= 1$$

(2) Information Gain

if a split is good, IG calculate how much entropy reduces after a split

$\rightarrow$ helps us choose feature that gives the highest IG.

$$\text{Information Gain} = \text{Entropy}_{\text{parent}} - \sum \frac{n_{\text{child}}}{n_{\text{total}}} \times \text{Entropy}_{\text{child}}$$

step: 1

calculate parent entropy

parent entropy (before split)

x_1 weather	x_2 Temperature	y play-Tennis
Sunny	Hot	NO
Sunny	Mild	NO
Cloudy	Hot	Yes
Rain	Mild	Yes
Rain	Cool	Yes
Cloudy	Cool	Yes
Rain	Hot	NO
Sunny	Cool	Yes

Class	Count	Prob
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Yes	5	$5/8 = 0.625$
-----	---	---------------

NO	3	$3/8 = 0.375$
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$$\text{Entropy}(\text{parent}) = -(0.625 \log_2 0.625 + 0.375 \log_2 0.375)$$

$$= 0.954$$

step 2: Try Each feature

weather

→ $\text{Entropy}(x_1)$

→ $\text{IG}(\text{parent-child})$

Temp

$\text{Entropy}(x_2)$

→ $\text{IG}(\text{parent-child})$

→ whichever is having the higher IG becomes the root node

Feature 1: weather

weather	count	Yes	NO	Entropy
Sunny	3	1	2	$-(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3}) = 0.918$
cloudy	2	2	0	0
Rain	3	2	1	0.918

x_1 weather	x_2 Temperature	y play-Tennis
Sunny	Hot	NO
Sunny	Mild	NO
Cloudy	Hot	Yes
Rain	Mild	Yes
Rain	Cool	Yes
Cloudy	Cool	Yes
Rain	Hot	NO
Sunny	Cool	Yes

weighted average entropy after split:

$$\frac{3}{8} \times 0.918 + \frac{2}{8} \times 0 + \frac{3}{8} \times 0.918 \rightarrow 0.688 \text{ (Entropy/Impurity)}$$

$$\text{Information Gain}(\text{weather}) \rightarrow 0.954 - 0.688$$

$$= 0.266$$

Feature: 2 $\rightarrow$ Temp

Temp	count	Yes	NO	Entropy
Hot	3	1	2	0.918
Mild	2	1	1	1
Cool	3	3	0	0

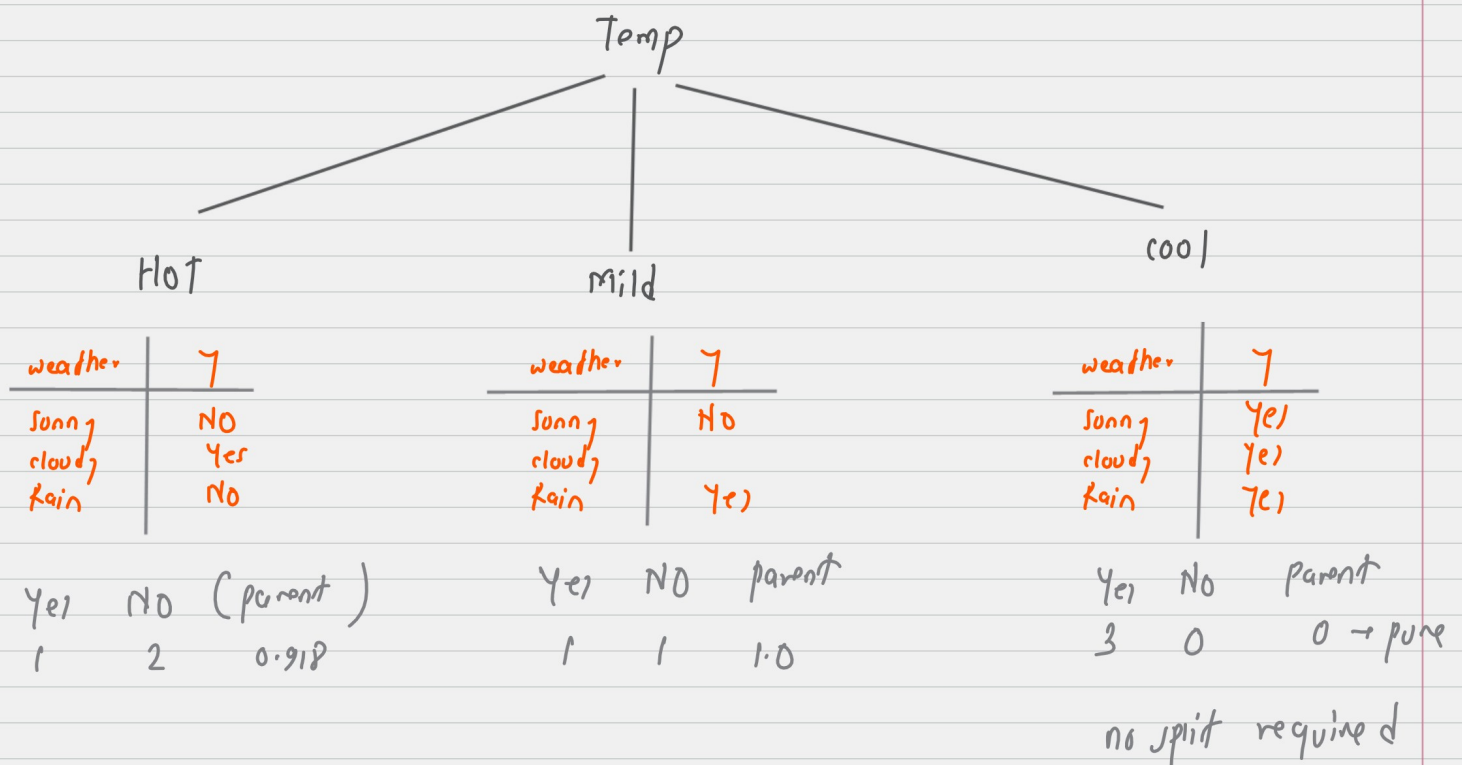
X1 weather	X2 Temperature	Y play-Tennis
Sunny	Hot	NO
Sunny	Mild	NO
cloudy	Hot	Yes
Rain	Mild	Yes
Rain	Cool	Yes
cloudy	Cool	Yes
Rain	Hot	NO
Sunny	Cool	Yes

weighted Entropy:

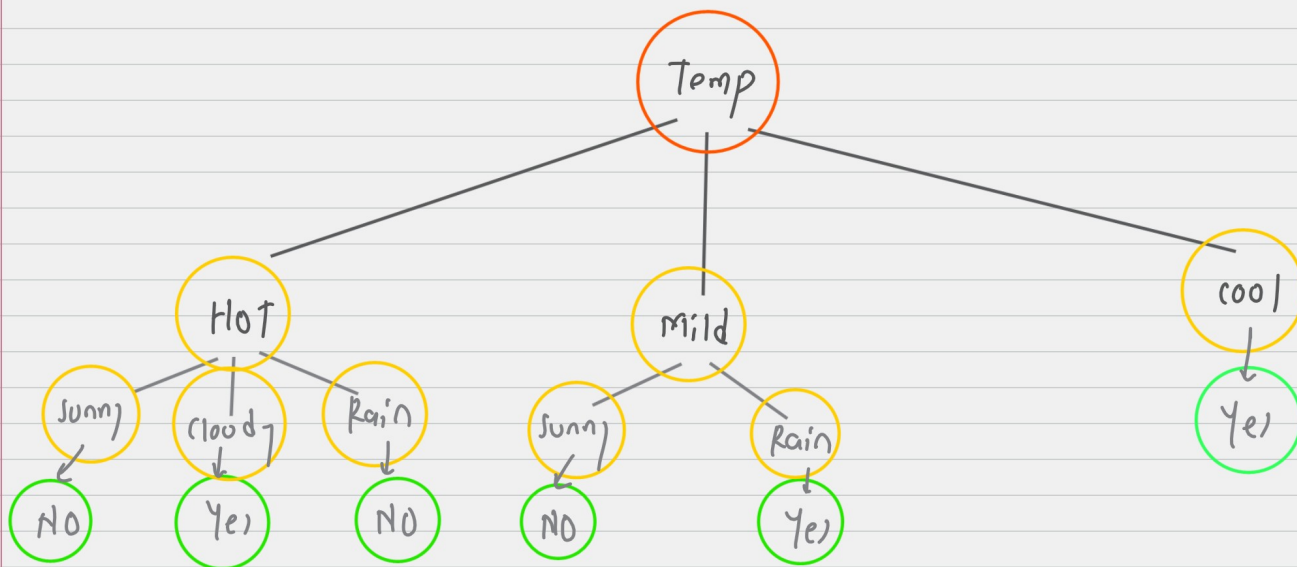
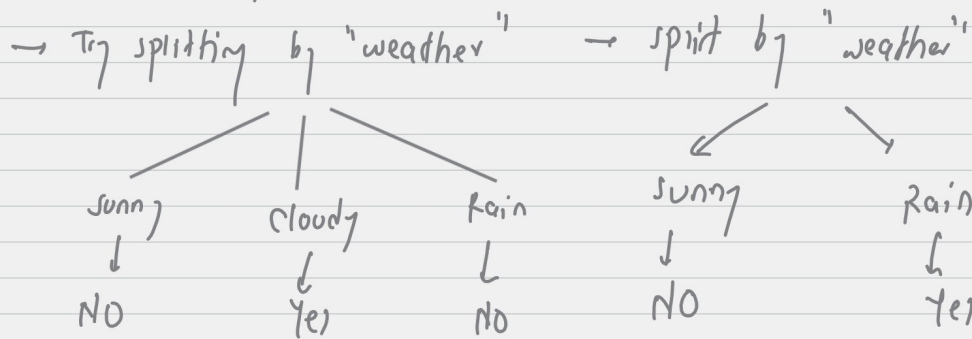
$$\frac{3}{8} \times 0.918 + \frac{2}{8} \times 1 + \frac{3}{8} \times 0 = 0.594$$

$$IG(\text{Temp}) = 0.954 - 0.594 = 0.360$$

$$\begin{array}{ccc} IG(\text{weather}) & & IG(\text{Temp}) \\ 0.268 & < & 0.360 \end{array}$$



→ we must split again to make it pure



unknown path follows majority voting

Decision - tree (A1yo)

→ compute parent Impurity
if found:

↳ compute Impurity of each feature split

↳ compute Information Gain

↳ pick the highest IG feature
at root

→ repeat recursively

→ when all nodes are pure, stop.

k-means clustering

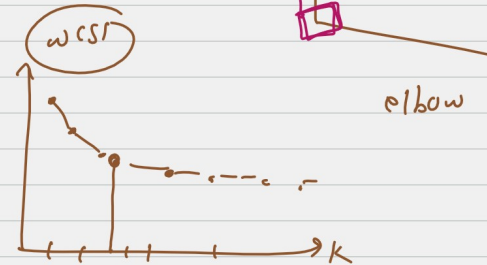
unsupervised model $\rightarrow k \rightarrow ?$



sizes $\sim ?$
 $k = 6$

$x_1, x_2, \dots, x_n$

$k \rightarrow 1 \rightarrow w(1) \rightarrow \text{highest}$
 $\rightarrow 2 \rightarrow w(2)$
 $\rightarrow 3 \rightarrow w(3)$
 $\rightarrow n \rightarrow w(n) \rightarrow \text{lowest}$

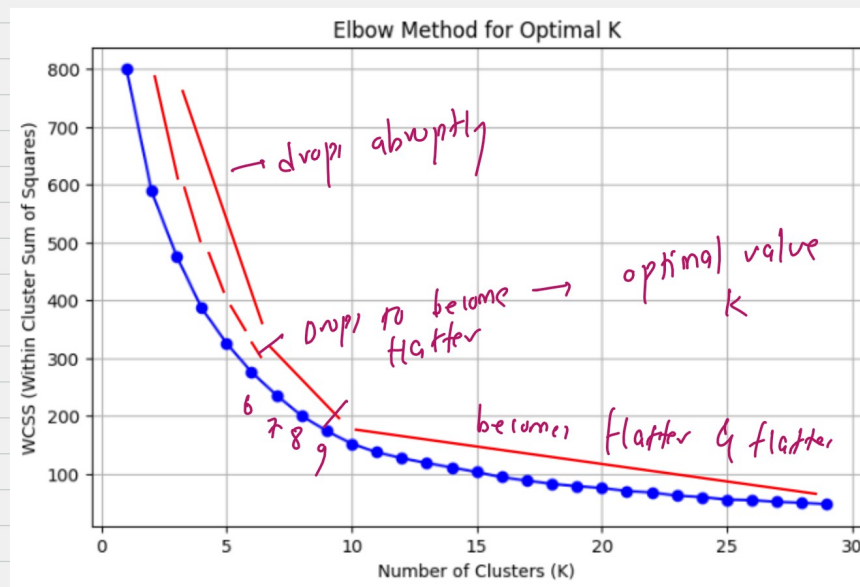


3-4 $\rightarrow$ SG $k \rightarrow 1-3 \rightarrow$ width

$n_{\text{init}} = 10$

$k = 1$

k-means++



n-features

PCA $\rightarrow$ dimensionality reduction algo
 (2D) $\rightarrow$ 2D features